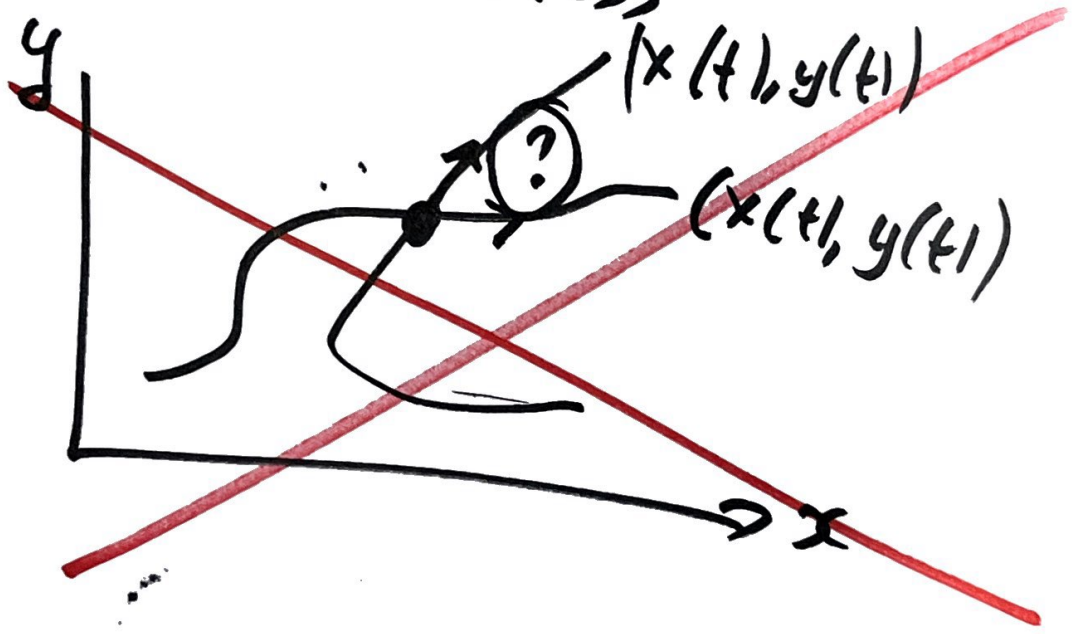
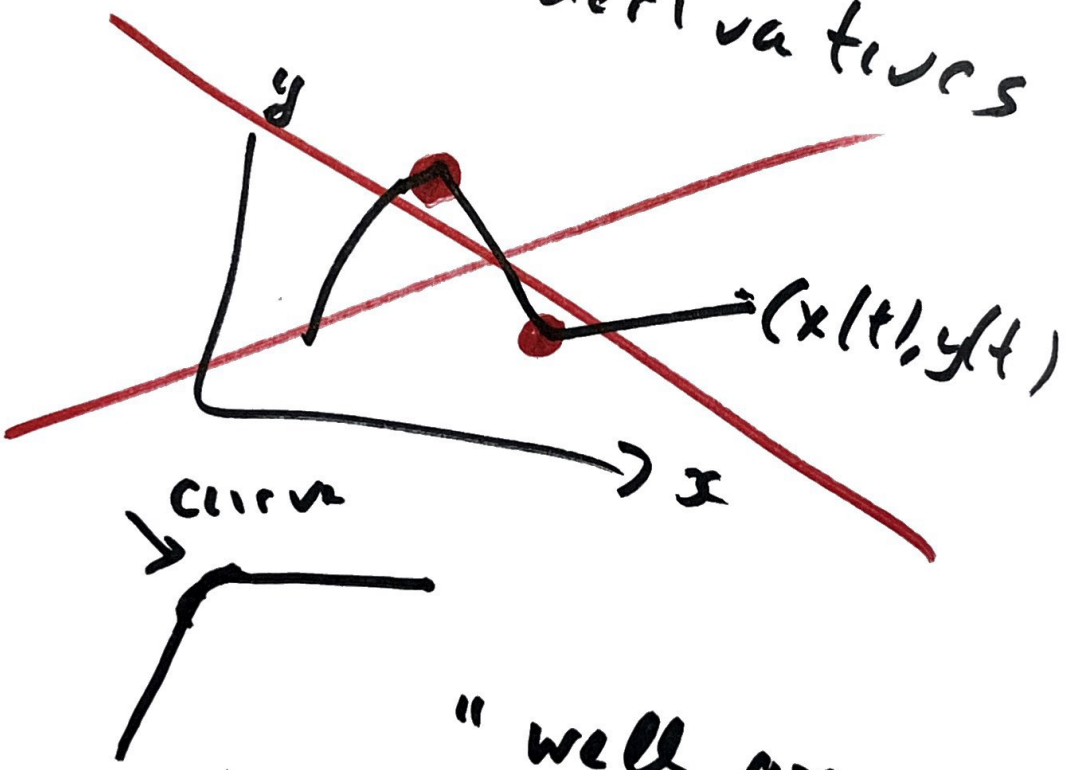
$$\left.\begin{aligned} \frac{d}{dt}x(t) &= f(x(t), y(t)) \\ \frac{d}{dt}y(t) &= g(x(t), y(t)) \\ y(t=a) &= y_0 \quad x(t=a) = x_0 \end{aligned}\right\}$$

If $f(x,y)$ and $g(x,y)$ are continuous with their first partial derivatives on an open connected set then solution to IVP $\exists!$

Phase trajectories on x-y plane do not cross



Since $f(x,y)$, $g(x,y)$ are continuous with their first derivatives

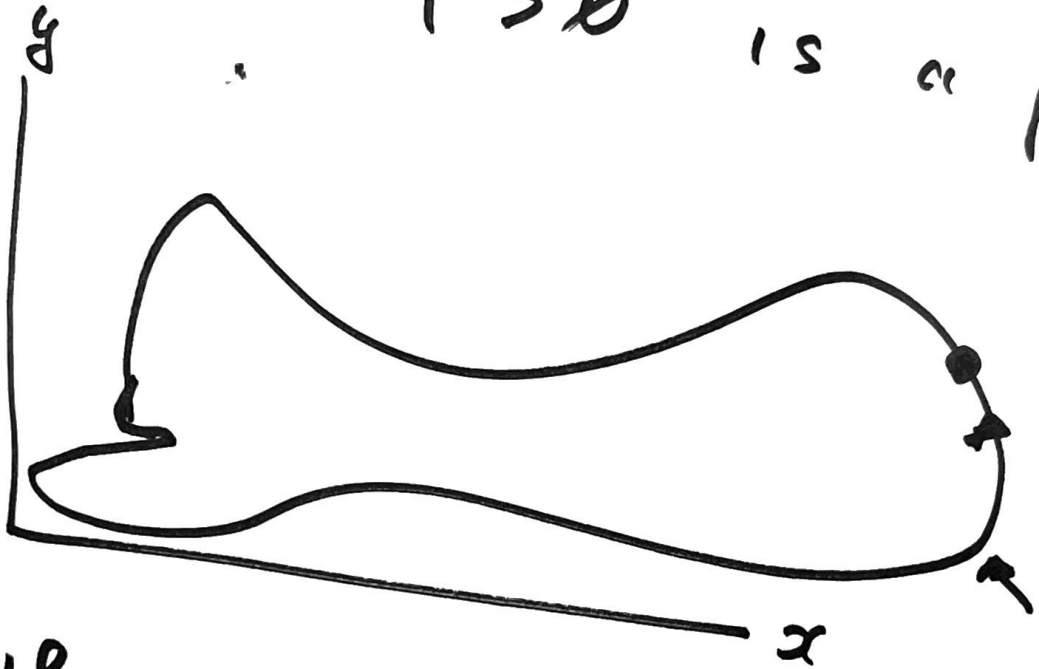


"well groomed trajectories"

periodic trajectory

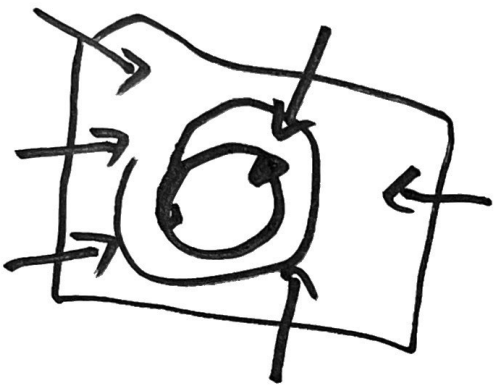
$$\underline{x}(t) = \underline{x}(t+T)$$

$T > 0$ is a period



If you start within periodic trajectory, you always will stay INSIDE periodic trajectory

Poincare Bendixson theorem

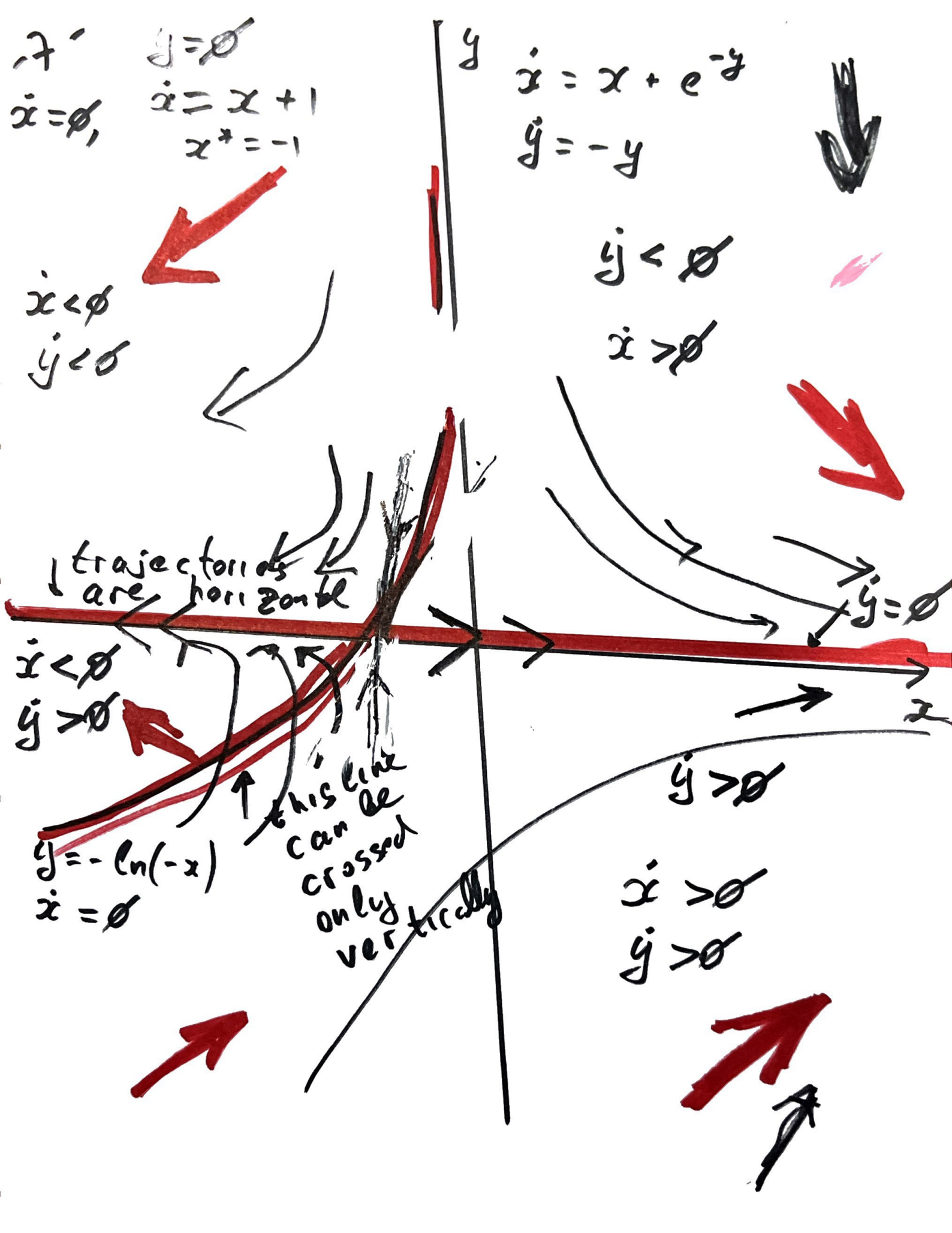


If there is an "attractive region" and there is no fixed points inside, then all trajectories will approach orbit

$$\left.\begin{aligned} \frac{d}{dt} x(t) &= x(t) + e^{-y(t)} \\ \frac{d}{dt} y(t) &= -y(t) \end{aligned}\right\}$$

Plot phase trajectories

Nullcline is a set of points where $\dot{x}(t) = 0$ OR $\dot{y}(t) = 0$


7.
$\dot{x} = 0$,
$\dot{y} = 0$
$\dot{x} = x + 1$
$x^* = -1$
$\dot{x} = x + e^{-y}$
$\dot{y} = -y$
y
$\dot{x} < 0$
$\dot{y} < 0$
$\dot{y} < 0$
$\dot{x} > 0$
trajectories are horizontal
$\dot{y} = 0$
$\dot{x} < 0$
$\dot{y} > 0$
x
$\dot{y} > 0$
this line can be crossed only vertically
$y = -\ln(-x)$
$\dot{x} = 0$
$\dot{x} > 0$
$\dot{y} > 0$

$\dot{y} = 0 \Rightarrow y = 0 \quad \dot{y} = -y$

$\dot{y} = 0 \Rightarrow$ trajectories are purely horizontal

$y > 0, \quad \dot{y} < 0$

above $y = 0$, $y > 0$, $\dot{y} < 0$, trajectories go down

below $y = 0$, $y < 0$, $\dot{y} > 0$, trajectories go up.

9-

$\dot{x} = 0 \Rightarrow x = -e^{-y}$

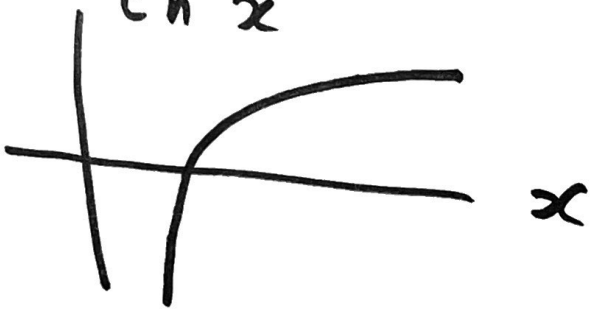
$\dot{x} = x + e^{-y}$

$-x = e^{-y}$

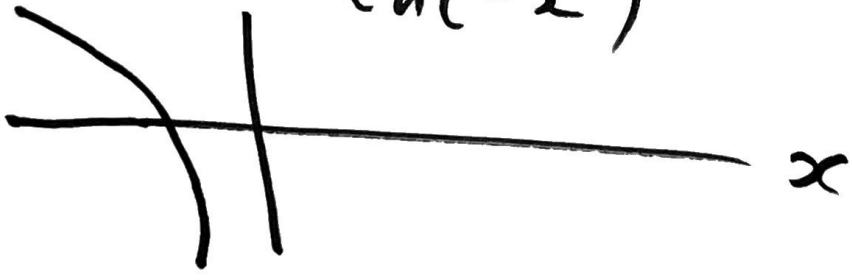
$\ln(-x) = -y$

$y = -\ln(-x)$

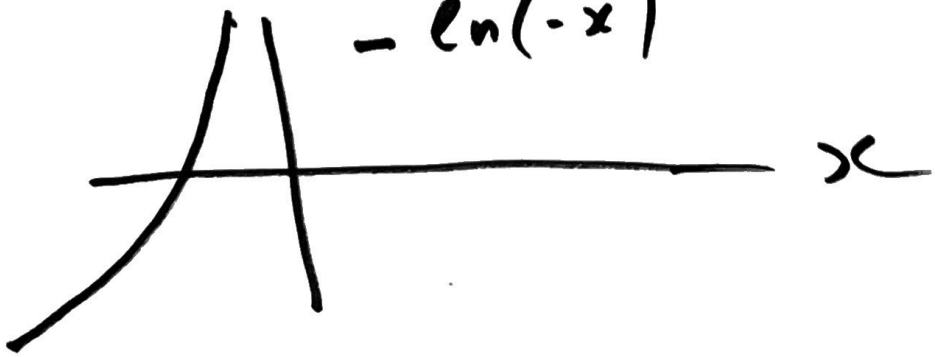
$\ln x$



$\ln(-x)$



$-\ln(-x)$



$\dot{x} = 0$ nullcline is $y = -\ln(-x)$ cross it only vertically

where two nullclines cross, there is a fixed point.

$y = x + e^{-y} = 0 \Rightarrow x = -1$

$\dot{x} =$ $\quad y = 0$

If x^*, y^* is a fixed point

$$x(t) = x^* + z(t)$$
$$y(t) = y^* + v(t)$$

$$\frac{d}{dt}\begin{pmatrix} z(t) \\ v(t) \end{pmatrix} = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{pmatrix} \begin{pmatrix} z(t) \\ v(t) \end{pmatrix}$$

(x^*, y^*)

$$\dot{x} = x + e^{-y}$$
$$\dot{y} = -y$$

$$\frac{d}{dt}\begin{pmatrix} z \\ v \end{pmatrix} = \begin{pmatrix} 1 & -e^{-y} \\ 0 & -1 \end{pmatrix} \begin{pmatrix} z \\ v \end{pmatrix}$$

$(x^* = -1, y^* = 0)$

$$J(-1, 0) = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}$$

$$\frac{d}{dt}\begin{pmatrix} z \\ v \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} z \\ v \end{pmatrix}$$

$$T = 0, \quad \Delta = -1$$

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4\Delta^2}}{2}$$

$$= \frac{0 \pm \sqrt{4}}{2} = \pm 1$$

$$\begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix} = \begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix} \qquad x_{12} = 0,$$

$$x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix} = -\begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix} \qquad x_{21} - x_{22} = -x_{21}$$

$$2x_{21} = x_{22}$$

$$x_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

-13-

$\dot{x} = -x + x^3$

$\dot{y} = 2y$

Sketch phase portrait, analyze fixed points.

$\dot{y} = 0 \Rightarrow y = 0$, $y > 0$, $\dot{y} > 0$

$y < 0$ $\dot{y} < 0$

$\dot{x} = 0 \Rightarrow \dot{x} = x(-1 + x^2)$

$\dot{x} = 0$ if $x = 1$,

$x = 0$

$x = -1$

$\dot{x} > 0$ IF

$x > 1$

$-1 < x < 0$

$\dot{x} = -x + x^3$



$\dot{x} < 0$

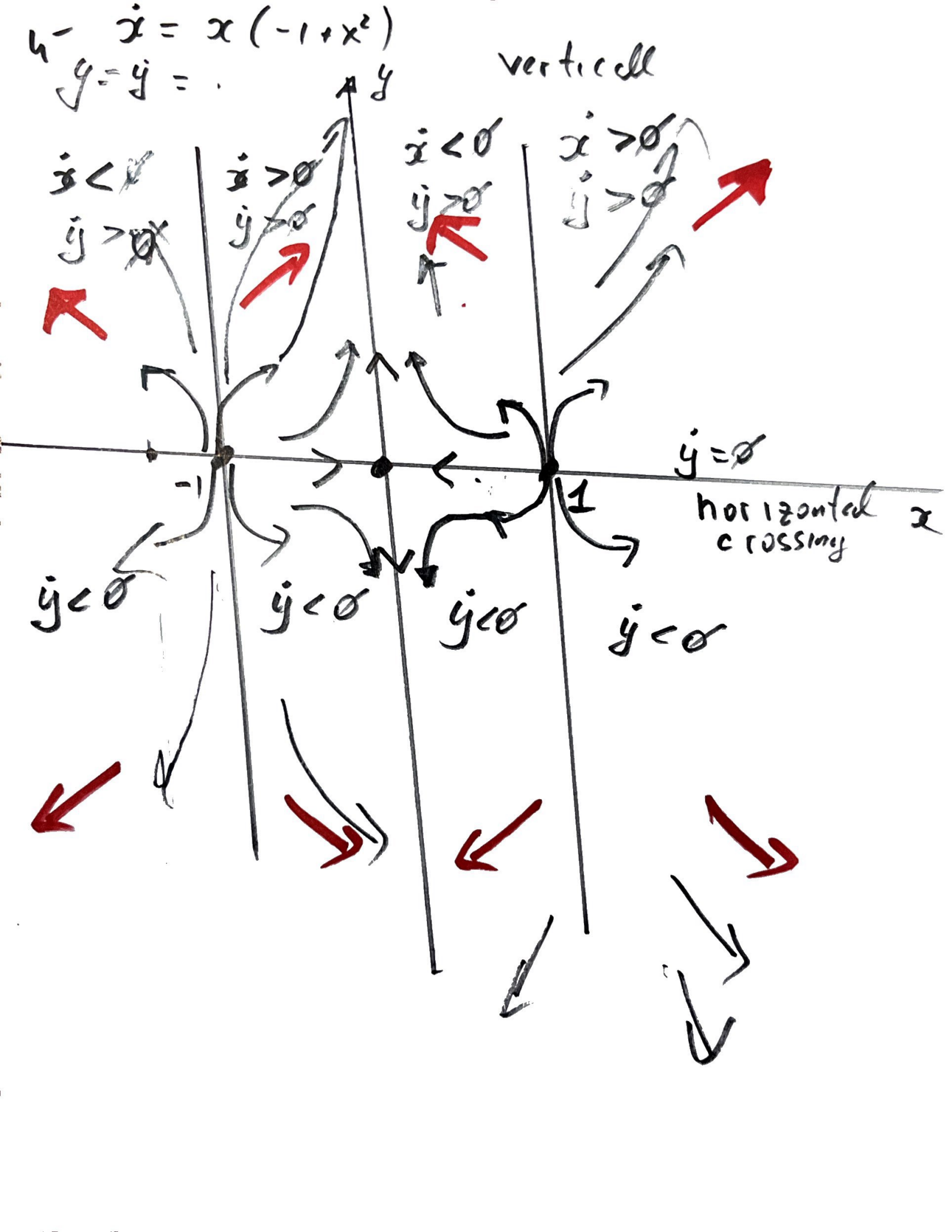
$x < -1$

$0 < x < 1$

4- $\dot{x} = x(-1+x^2)$

$\dot{y} = y$

vertical



$$\dot{x} = f(x,y) = -x + x^3$$
$$\dot{y} = g(x,y) = y;$$

$$J = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{pmatrix} =$$

$$= \begin{pmatrix} -1+3x^2 & 0 \\ 0 & 1 \end{pmatrix}$$

$(0,0)$
$(1,0)$
$(-1,0)$

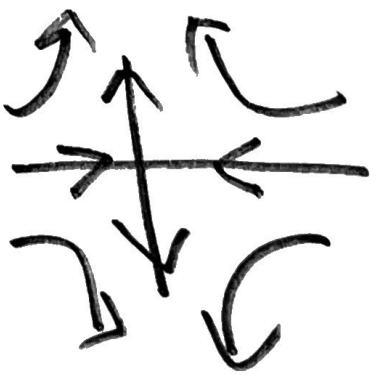
$(0,0)$

$$\dot{x} = x(-1+x^2)$$
$$\dot{y} = y$$

$$J(0,0) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\lambda_1 = -1;\ x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

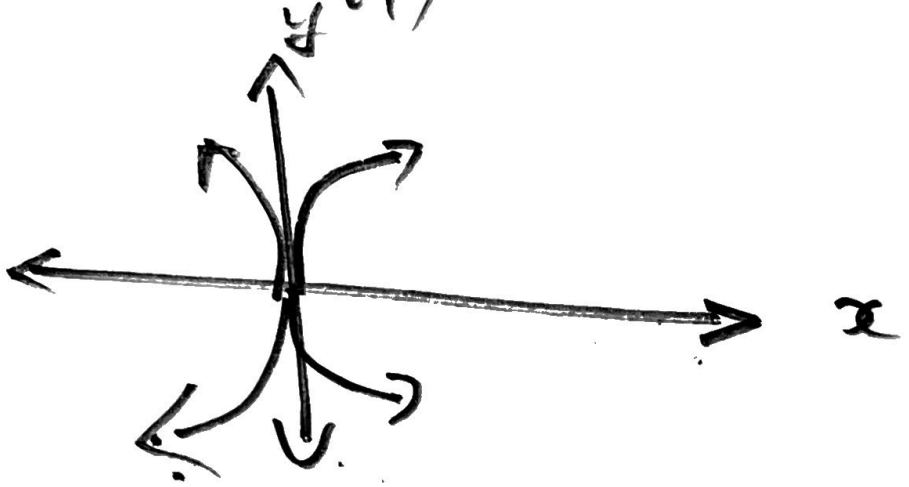
$\lambda_2 = 1;\ x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$



16 $T(1,0) = \begin{pmatrix} -1+3 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

$\lambda_1 = 2;\ \underline{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\lambda_2 = 1;\ x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$



17

$$\dot{x} = -y + \alpha \times (x^2+y^2)$$

$$\dot{y} = x + \alpha\, y(x^2+y^2)$$

$\alpha > 0$ is a given, fixed parameter

Fixed point is

$(x^*, y^*) = (0,0)$:

$$J(x,y) = \begin{pmatrix} 3x^2\cdot\alpha + \alpha y^2 & -1 + 2\alpha xy \\ 1 + 2\alpha xy & \alpha x^2 + 3\alpha y^2 \end{pmatrix}$$

$$J(0,0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

center

18

$$x = r\cos\varphi$$

$$y = r\sin\varphi$$

$$r = \sqrt{x^2 + y^2}$$

$$\tan\varphi = y/x$$

$$\varphi = \text{Arc Tan}\left(\frac{y}{x}\right)$$

$$\frac{D}{Dx}\,\text{Arctan}\,X = \frac{1}{1+x^2}$$